

# STABLE MATCHING AND A CONDITIONAL TWO-TO-ONE MAP FOR TUNNELL'S TERNARY FORMS

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ABSTRACT. For an odd positive squarefree integer  $n$ , let  $\mathcal{B}_n$  and  $\mathcal{A}_n$  be the representation sets of  $n$  by  $2x^2 + y^2 + 8z^2$  and  $2x^2 + y^2 + 32z^2$ , respectively. Assuming  $|\mathcal{B}_n| = 2|\mathcal{A}_n|$ , we construct a deterministic map  $\Phi_n : \mathcal{B}_n \rightarrow \mathcal{A}_n$  whose fibers have cardinality two. Halving the third coordinate gives the even branch. On the odd branch, three integral quarter-turns define a partial bijection. We complete it by passing the residual points to their antipodal orbits and applying stable matching, with preferences determined by primitive midpoint directions. Equal-rank edges are disjoint, so the stable matching is unique. Its preference lists can be generated arithmetically from rank-two lattice cosets, and the proposal counts form a parking function with total at most  $m(m+1)/2$ , where  $m$  is the number of residual orbits on either side. For odd squarefree congruent numbers, Tunnell's theorem supplies the cardinality hypothesis. The construction assumes, rather than proves, this identity.

## 1. INTRODUCTION

Equalities between finite representation numbers naturally raise a finer combinatorial question: can the individual representations be matched in a structured way? We study this question for the ternary quadratic forms that occur in Tunnell's work on congruent numbers. For an odd positive integer  $n$ , let

$$\begin{aligned} (1) \quad \mathcal{B}_n &= \{(x, y, z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 8z^2 = n\}, \\ (2) \quad \mathcal{A}_n &= \{(u, v, w) \in \mathbb{Z}^3 : 2u^2 + v^2 + 32w^2 = n\}. \end{aligned}$$

A ternary quadratic form is a homogeneous quadratic polynomial in three variables, and a representation of  $n$  is simply an integer triple at which the form has value  $n$ . We call  $n$  *squarefree* if no prime square divides it. Both sets above are finite because each squared coordinate is bounded by  $n$ .

A positive integer is *congruent* if it is the area of a right triangle with rational side lengths. Equivalently, the elliptic curve  $y^2 = x^3 - n^2x$  has a rational point of infinite order; see [6, Chap. I, §9] and [9, pp. 323–324].

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A consequence of Tunnell's theorem is that every odd positive squarefree congruent number satisfies

$$(3) \quad |\mathcal{B}_n| = 2|\mathcal{A}_n|.$$

This consequence is the only input from Tunnell's theorem used below; no modular-form or elliptic-curve argument enters the construction itself.

*Example 1.1* (The balance for  $n = 41$ ). The equations themselves make this first example finite and elementary. For  $n = 41$ , a solution in  $\mathcal{B}_{41}$  must satisfy  $|x| \leq 4$ ,  $|y| \leq 6$ , and  $|z| \leq 2$ ; a solution in  $\mathcal{A}_{41}$  must satisfy the same first two bounds and  $|w| \leq 1$ . Sorting by the last coordinate gives

| $\mathcal{B}_{41} : z$ | 0 | 1 | -1 | 2 | -2 | total |
|------------------------|---|---|----|---|----|-------|
| number of $(x, y)$     | 4 | 8 | 8  | 6 | 6  | 32    |

and

| $\mathcal{A}_{41} : w$ | 0 | 1 | -1 | total |
|------------------------|---|---|----|-------|
| number of $(u, v)$     | 4 | 6 | 6  | 16    |

Thus the required 2:1 balance holds. For a single visible instance,

$$(2, -1, 2) \in \mathcal{B}_{41}, \quad (2, -1, 1) \in \mathcal{A}_{41},$$

and the even-branch map simply halves the last coordinate.

Methods such as those of Bach and Ryan [2] compute the representation numbers in this identity. Our purpose is different. We take (3) as an explicit hypothesis and seek a structured map between the representations that it counts. The construction therefore neither proves Tunnell's identity nor supplies a new congruent-number criterion.

Separate  $\mathcal{B}_n$  according to the parity of its third coordinate:

$$\begin{aligned} \mathcal{B}_n^{\text{even}} &= \{(x, y, z) \in \mathcal{B}_n : z \equiv 0 \pmod{2}\}, \\ \mathcal{B}_n^{\text{odd}} &= \{(x, y, z) \in \mathcal{B}_n : z \equiv 1 \pmod{2}\}. \end{aligned}$$

The map

$$(4) \quad (x, y, 2w) \mapsto (x, y, w)$$

is a bijection  $\mathcal{B}_n^{\text{even}} \xrightarrow{\sim} \mathcal{A}_n$ . Consequently (3) is equivalent to  $|\mathcal{B}_n^{\text{odd}}| = |\mathcal{A}_n|$ , and the essential problem is to construct a bijection from the odd branch to the target set.

The residual problem is naturally bipartite. After a direct partial bijection is removed and the remaining points are passed to their  $\{\pm 1\}$ -orbits, every source orbit is joined to every target orbit. A primitive midpoint direction assigns a rank to each edge. Both endpoints order their incident edges by this rank, while equal-rank edges are vertex-disjoint. This is an instance of stable marriage with globally ranked pairs in the sense of Abraham, Levavi, Manlove, and O'Malley [1]. Its special edge structure forces a unique stable matching.

Stable matching and parking functions are classical combinatorial objects [4, 5, 7, 8]. Here they arise from the arithmetic geometry of two representation sets. The direct edges come from integral orthogonal transformations, whereas the residual edge ranking comes from primitive midpoint lines. Although balance alone would permit an arbitrary pairing, these arithmetic choices determine the map after the orientation conventions below have been fixed. The construction is equivariant under the antipodal involution  $X \mapsto -X$ , and the same edge data determine its inverse.

**Theorem 1.2** (Conditional two-to-one map). *Let  $n$  be an odd positive squarefree integer satisfying  $|\mathcal{B}_n| = 2|\mathcal{A}_n|$ . Fix the orientation and ordering conventions introduced below. Then the construction defines a deterministic map*

$$\Phi_n : \mathcal{B}_n \longrightarrow \mathcal{A}_n$$

*such that every element of  $\mathcal{A}_n$  has exactly two preimages. On  $\mathcal{B}_n^{\text{even}}$ , the map is (4). On  $\mathcal{B}_n^{\text{odd}}$ , it is the disjoint union of three quarter-turn branches and the unique stable matching of the remaining orbit sets. The later arithmetic generator and deferred-acceptance procedure compute this matching without first storing its complete edge table.*

Here a *fiber* is the set of all source points mapping to one target. The architecture of the theorem is the following. The top row gives one preimage of every target, and the bottom row gives the other:

$$(5) \quad \begin{array}{c} \mathcal{B}_n^{\text{even}} \xrightarrow[\text{bijection}]{(x,y,2w) \mapsto (x,y,w)} \mathcal{A}_n, \\ \mathcal{B}_n^{\text{odd}} = D_n \sqcup \mathcal{B}_{n,o}^{\text{odd}} \xrightarrow[\text{bijection}]{\Lambda_n \sqcup \Psi_n^\circ} I_n \sqcup \mathcal{A}_{n,o} = \mathcal{A}_n. \end{array}$$

The residual arrow in the bottom row is built through

$$(6) \quad \mathcal{B}_{n,o}^{\text{odd}} \xrightarrow{\iota_B} \mathcal{S}_n^\circ \longrightarrow \overline{\mathcal{S}}_n \xrightarrow{\overline{M}_n} \overline{\mathcal{T}}_n \longrightarrow \mathcal{T}_n^\circ \xrightarrow{\iota_A^{-1}} \mathcal{A}_{n,o}.$$

The unlabeled arrows pass to antipodal orbits and then lift back with a specified sign. The cardinality hypothesis is used to prove that  $\overline{M}_n$  is perfect; it is not used to choose the edge ranks.

**Corollary 1.3.** *Theorem 1.2 applies to every odd positive squarefree congruent number  $n$ .*

Section 2 introduces the two coordinate models, and Section 3 constructs the direct branches. Section 4 constructs the unique residual matching, and Section 5 immediately assembles the two-to-one map. Sections 6–9 then show how to compute the residual matching and bound its proposal phase. We finish with the full  $n = 41$  residual table and two further questions.

The examples at  $n = 41$  follow two complementary lanes, not one point through both odd-branch mechanisms:

| lane                | odd source    | target       | even source    |
|---------------------|---------------|--------------|----------------|
| direct quarter-turn | $(-4, -1, 1)$ | $(2, -1, 1)$ | $(2, -1, 2)$   |
| residual matching   | $(-2, -5, 1)$ | $(4, -3, 0)$ | $(4, -3, 0)$ . |

Each lane therefore ends in a complete two-element fiber, but the two odd mechanisms act on disjoint subsets.

## 2. COORDINATE MODELS FOR THE DIRECT AND RESIDUAL CONSTRUCTIONS

We use one coordinate model for the direct local branches and another for midpoint directions. They are parallel descriptions of the original sets, not successive changes of coordinates:

$$\begin{array}{ccccc}
 \mathcal{B}_n^{\text{odd}} & \xleftarrow[e_B^{-1}]{e_B} & L_1(n) & \xrightarrow{J_1, J_2, J_3} & L_0(n) \xleftarrow[e_A]{e_A^{-1}} \mathcal{A}_n, \\
 \mathcal{B}_n^{\text{odd}} & \xleftarrow[\iota_B^{-1}]{\iota_B} & \mathcal{S}_n & \xleftrightarrow{\text{midpoint lines}} & \mathcal{T}_n \xleftarrow[\iota_A]{\iota_A^{-1}} \mathcal{A}_n.
 \end{array}$$

The first row makes quarter-turns integral. The second puts sources and targets on the same quadratic surface so that their half-sums and half-differences are integral.

For reference, the main set and map symbols used below are collected here:

|                                                                           |                                                                                          |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| $D_n, I_n, \Lambda_n$                                                     | direct odd-source domain, its target image, and the quarter-turn bijection between them; |
| $\mathcal{B}_{n,\circ}^{\text{odd}}, \mathcal{A}_{n,\circ}, \Psi_n^\circ$ | points left after the direct layer and their residual bijection;                         |
| $\mathcal{S}_n, \mathcal{T}_n$                                            | midpoint-coordinate lifts of the odd source and target sets;                             |
| $\bar{\mathcal{S}}_n, \bar{\mathcal{T}}_n$                                | antipodal residual orbit sets;                                                           |
| $\mathcal{P}_Q$                                                           | canonically oriented primitive midpoint directions;                                      |
| $\Psi_n, \Phi_n$                                                          | the full odd-branch bijection and the final two-to-one map.                              |

**2.1. Sum-of-three-squares coordinates for the direct map.** For  $\epsilon \in \{0, 1\}$ , put

$$L_\epsilon(n) = \{(a, b, c) \in \mathbb{Z}^3 : a^2 + b^2 + c^2 = n, \quad a \equiv b \pmod{4}, \quad \frac{a-b}{4} \equiv \epsilon \pmod{2}\}.$$

**Lemma 2.1** (Sum-of-three-squares coordinates). *The maps*

$$e_B(x, y, z) = (x + 2z, x - 2z, y), \quad e_A(u, v, w) = (u + 4w, u - 4w, v)$$

are bijections

$$e_B : \mathcal{B}_n^{\text{odd}} \xrightarrow{\sim} L_1(n), \quad e_A : \mathcal{A}_n \xrightarrow{\sim} L_0(n),$$

with inverses

$$\begin{aligned} e_B^{-1}(a, b, c) &= \left( \frac{a+b}{2}, c, \frac{a-b}{4} \right), \\ e_A^{-1}(a, b, c) &= \left( \frac{a+b}{2}, c, \frac{a-b}{8} \right). \end{aligned}$$

*Proof.* The norm identities are immediate:

$$\begin{aligned} (x+2z)^2 + (x-2z)^2 + y^2 &= 2x^2 + y^2 + 8z^2, \\ (u+4w)^2 + (u-4w)^2 + v^2 &= 2u^2 + v^2 + 32w^2. \end{aligned}$$

The congruence conditions record whether  $(a-b)/4$  is odd or even. Conversely, if  $(a, b, c) \in L_1(n)$ , then  $(a+b)/2$  and  $(a-b)/4$  are integral, the latter is odd, and the first norm identity gives a point of  $\mathcal{B}_n^{\text{odd}}$ . If  $(a, b, c) \in L_0(n)$ , then  $(a-b)/8$  is integral and the second identity gives a point of  $\mathcal{A}_n$ . The displayed inverse formulas therefore give inverses on both sides.  $\square$

*Example 2.2* (The two coordinate changes at  $n = 41$ ). Consider the odd source  $X = (-4, -1, 1) \in \mathcal{B}_{41}^{\text{odd}}$ . Lemma 2.1 sends it to

$$e_B(X) = (-2, -6, -1), \quad (-2)^2 + (-6)^2 + (-1)^2 = 41.$$

Here  $(-2) - (-6) = 4$ , so  $(a-b)/4 = 1$  is odd, as required for  $L_1(41)$ . Likewise, for  $Y = (2, -1, 1) \in \mathcal{A}_{41}$ ,

$$e_A(Y) = (6, -2, -1), \quad 6^2 + (-2)^2 + (-1)^2 = 41,$$

and  $(6 - (-2))/4 = 2$  is even, as required for  $L_0(41)$ . The inverse formulas recover  $X$  and  $Y$  from these triples.

## 2.2. Midpoint coordinates for the residual graph. Put

$$Q(X_1, X_2, X_3) = 2X_1^2 + X_2^2 + 2X_3^2$$

and write

$$\langle X, Y \rangle_Q = 2X_1Y_1 + X_2Y_2 + 2X_3Y_3$$

for its associated bilinear form. Thus  $Q(X+Y) = Q(X) + 2\langle X, Y \rangle_Q + Q(Y)$ . Define

$$\begin{aligned} \mathcal{S}_n &= \{E \in \mathbb{Z}^3 : Q(E) = n, E_3 \equiv 2 \pmod{4}\}, \\ \mathcal{T}_n &= \{F \in \mathbb{Z}^3 : Q(F) = n, F_3 \equiv 0 \pmod{4}\}. \end{aligned}$$

The lifts

$$\iota_B(x, y, z) = (x, y, 2z), \quad \iota_A(u, v, w) = (u, v, 4w)$$

identify  $\mathcal{B}_n^{\text{odd}}$  with  $\mathcal{S}_n$  and  $\mathcal{A}_n$  with  $\mathcal{T}_n$ .

**Lemma 2.3** (Midpoint parity). *If  $E \in \mathcal{S}_n$  and  $F \in \mathcal{T}_n$ , then*

$$\frac{E+F}{2}, \quad \frac{E-F}{2}$$

*are integral vectors with odd third coordinate.*

*Proof.* Since  $n$  is odd, the second coordinates of both  $E$  and  $F$  are odd. Their third coordinates are congruent to 2 and 0 modulo 4, respectively. Consequently, modulo 8 one has

$$n \equiv 2E_1^2 + 1 \equiv 2F_1^2 + 1,$$

so  $E_1$  and  $F_1$  have the same parity. The second-coordinate halves are integral because both entries are odd, and the third-coordinate halves are odd because  $E_3 \equiv 2 \pmod{4}$  and  $F_3 \equiv 0 \pmod{4}$ . Thus both displayed vectors are integral and have odd third coordinate.  $\square$

For  $0 \neq v \in \mathbb{Z}^3$ , define

$$\text{cont}(v) = \gcd(|v_1|, |v_2|, |v_3|).$$

Call  $v$  *primitive* if  $\text{cont}(v) = 1$ . Let  $\mathcal{P}_Q$  contain the unique representative, with first nonzero coordinate positive, of every pair  $\{d, -d\}$  of primitive vectors with  $d_3$  odd. Order  $\mathcal{P}_Q$  lexicographically by

$$(7) \quad \text{key}(d) = (Q(d), d_1, d_2, d_3).$$

This direction convention differs from the point-orbit convention used later: directions have first nonzero coordinate positive, whereas an antipodal point orbit is represented by its lexicographically smaller point. Keeping the two rules explicit prevents their signs from being confused.

**Lemma 2.4** (Primitive generator of the midpoint line). *For every  $E \in \mathcal{S}_n$  and  $F \in \mathcal{T}_n$ , there are unique  $d \in \mathcal{P}_Q$  and a nonzero odd integer  $q$  such that*

$$\frac{E + F}{2} = qd.$$

*They satisfy*

$$\langle E, d \rangle_Q = qQ(d), \quad F = -E + 2qd.$$

*For a fixed  $d$ , the edges  $E \leftrightarrow F$  obtained in this way are pairwise disjoint.*

*Proof.* By Lemma 2.3, the midpoint is a nonzero integral vector with odd third coordinate. Dividing by its content and applying the orientation convention gives  $q$  and  $d$ , with  $q$  odd. If  $h = (E - F)/2$ , equality of the norms gives  $\langle d, h \rangle_Q = 0$ , and hence  $\langle E, d \rangle_Q = qQ(d)$  and  $F = -E + 2qd$ . Uniqueness follows from the unique sign-normalized primitive generator of  $\mathbb{Q}(E + F) \cap \mathbb{Z}^3$ , the integer points on the rational line spanned by  $E + F$ . For fixed  $E$  and  $d$ , the identity  $q = \langle E, d \rangle_Q / Q(d)$  determines  $q$ , and then the formula determines  $F$ . Interchanging the roles of the endpoints gives the same conclusion at a target. Hence the edges belonging to a fixed direction are pairwise disjoint.  $\square$

*Example 2.5* (A midpoint direction at  $n = 41$ ). Take the lifted residual points

$$E = (-2, -5, 2) \in \mathcal{S}_{41}, \quad F = (4, -3, 0) \in \mathcal{T}_{41}.$$

Both have  $Q$ -value 41. Their half-sum and half-difference are

$$\frac{E + F}{2} = (1, -4, 1), \quad \frac{E - F}{2} = (-3, -1, 1),$$

so Lemma 2.3 is visible directly: both vectors are integral and have odd third coordinate. The first vector is already primitive and canonically oriented, hence  $q = 1$  and  $d = (1, -4, 1)$ . Moreover,

$$Q(d) = 20, \quad \langle E, d \rangle_Q = 20 = qQ(d), \quad -E + 2d = F.$$

The other midpoint line is generated, after sign normalization, by  $e = (3, 1, -1)$ , whose key begins with  $Q(e) = 21$ . Thus the key of  $d$  is the smaller of the two, explaining which representative-level edge is preferred when the construction later passes to antipodal orbits. Thus the midpoint line determines this edge exactly as in Lemma 2.4.

### 3. A DIRECT QUARTER-TURN CORRESPONDENCE

Define three integral rotations in the sum-of-three-squares model by

$$J_1(a, b, c) = (a, -c, b), \quad J_2(a, b, c) = (c, b, -a), \quad J_3(a, b, c) = (-b, a, c).$$

Each is an element of  $\mathrm{SO}_3(\mathbb{Z})$ : it is an integer matrix that preserves squared length and orientation. Geometrically,  $J_i$  fixes one coordinate axis and rotates the other two coordinates through a right angle.

**Proposition 3.1** (Quarter-turn domains). *Let  $(a, b, c) \in L_1(n)$ . Then*

$$\begin{aligned} J_1(a, b, c) \in L_0(n) &\iff a + c \equiv 0 \pmod{8}, \\ J_2(a, b, c) \in L_0(n) &\iff c - b \equiv 0 \pmod{8}, \\ J_3(a, b, c) \in L_0(n) &\iff a + b \equiv 0 \pmod{8}. \end{aligned}$$

*At most one of these three congruences holds.*

*Proof.* The differences between the first two coordinates of the three images are  $a + c$ ,  $c - b$ , and  $-(a + b)$ , respectively. Membership in  $L_0(n)$  is therefore equivalent to the stated divisibility by 8.

In original odd-source coordinates,  $c = y$  is odd and  $a - b = 4z \equiv 4 \pmod{8}$ . If any two of the displayed congruences held, then, after eliminating  $a$  or  $b$ , one would obtain

$$a - b \equiv -2c \pmod{8}.$$

Since  $c$  is odd, the right-hand side is 2 or 6 modulo 8, contradicting  $a - b \equiv 4 \pmod{8}$ . Hence the three domains are pairwise disjoint.  $\square$

Transferring the rotations back to the original coordinates gives the following partial map. For  $(x, y, z) \in \mathcal{B}_n^{\mathrm{odd}}$  with  $x + 2z + y \equiv 0 \pmod{8}$ , set

$$(9) \quad \Lambda_1(x, y, z) = \left( \frac{x + 2z - y}{2}, x - 2z, \frac{x + 2z + y}{8} \right).$$

If  $y - x + 2z \equiv 0 \pmod{8}$ , set

$$(9) \quad \Lambda_2(x, y, z) = \left( \frac{x - 2z + y}{2}, -x - 2z, \frac{y - x + 2z}{8} \right).$$

Finally, if  $x \equiv 0 \pmod{4}$ , set

$$(10) \quad \Lambda_3(x, y, z) = \left( 2z, y, -\frac{x}{4} \right).$$

The domains are pairwise disjoint by Proposition 3.1.

**Theorem 3.2** (Direct partial bijection). *The formulas (8)–(10) define a partial bijection*

$$\Lambda_n : D_n \xrightarrow{\sim} I_n$$

from a subset  $D_n \subseteq \mathcal{B}_n^{\text{odd}}$  to a subset  $I_n \subseteq \mathcal{A}_n$ . Its inverse is determined from a target  $(u, v, w) \in \mathcal{A}_n$  by the following mutually exclusive conditions and formulas:

$$\begin{aligned} u + 4w - v \equiv 4 \pmod{8} : & \quad \left( \frac{u + 4w + v}{2}, -u + 4w, \frac{u + 4w - v}{4} \right), \\ u - 4w + v \equiv 4 \pmod{8} : & \quad \left( \frac{u - 4w - v}{2}, u + 4w, \frac{4w - u - v}{4} \right), \\ u \equiv 2 \pmod{4} : & \quad \left( -4w, v, \frac{u}{2} \right). \end{aligned}$$

Moreover,  $\Lambda_n$  is equivariant under the antipodal involution  $X \mapsto -X$ .

*Proof.* The construction is  $e_A^{-1} J_i e_B$  on each branch, so norm preservation and integrality follow from Proposition 3.1. The inverse formulas are  $e_B^{-1} J_i^{-1} e_A$ . Their conditions state exactly that the inverse image lies in  $L_1(n)$ . They are mutually exclusive: the first two cannot both hold because  $v$  is odd, and either of them is incompatible with  $u \equiv 2 \pmod{4}$  by parity. Direct composition gives the two identity maps. Negation equivariance follows because each  $J_i$  is linear and each domain predicate is sign invariant.  $\square$

*Example 3.3* (The three quarter-turn branches). The first two branches already occur at  $n = 11$ ; the direct construction does not require the cardinality hypothesis. Direct substitution gives

| branch      | $X$           | $e_B(X)$      | $\Lambda_i(X)$ |
|-------------|---------------|---------------|----------------|
| $\Lambda_1$ | $(-1, -1, 1)$ | $(1, -3, -1)$ | $(1, -3, 0)$   |
| $\Lambda_2$ | $(-1, 1, -1)$ | $(-3, 1, 1)$  | $(1, 3, 0)$ .  |

For the first row,  $a + c = 1 - 1 = 0$ , so  $J_1$  applies and  $J_1(1, -3, -1) = (1, 1, -3) = e_A(1, -3, 0)$ . For the second row,  $c - b = 1 - 1 = 0$ , so  $J_2$  applies and  $J_2(-3, 1, 1) = (1, 1, 3) = e_A(1, 3, 0)$ . These calculations exhibit the first two domain tests and their images.

For the  $n = 41$  direct lane, continue with  $X = (-4, -1, 1)$  from Example 2.2. Its sum-of-three-squares coordinates are  $(a, b, c) = (-2, -6, -1)$ . Since

$$a + b = -8 \equiv 0 \pmod{8},$$

Proposition 3.1 selects  $J_3$  and neither of the other two quarter-turns. Indeed,

$$J_3(-2, -6, -1) = (6, -2, -1) = e_A(2, -1, 1).$$



In the original coordinates this is exactly

$$\Lambda_3(-4, -1, 1) = (2, -1, 1).$$

The inverse condition is  $u \equiv 2 \pmod{4}$ , and the third inverse formula gives  $(-4w, v, u/2) = (-4, -1, 1)$ .

Set

$$\mathcal{B}_{n,\circ}^{\text{odd}} = \mathcal{B}_n^{\text{odd}} \setminus D_n, \quad \mathcal{A}_{n,\circ} = \mathcal{A}_n \setminus I_n,$$

and let  $\mathcal{S}_n^\circ = \iota_B(\mathcal{B}_{n,\circ}^{\text{odd}})$  and  $\mathcal{T}_n^\circ = \iota_A(\mathcal{A}_{n,\circ})$ . Under the hypothesis of Theorem 1.2,

$$(11) \quad |\mathcal{S}_n^\circ| = |\mathcal{T}_n^\circ|.$$

Indeed,  $|\mathcal{B}_n^{\text{odd}}| = |\mathcal{A}_n|$  and  $\Lambda_n$  removes equally many points from the two sides.

#### 4. GLOBALLY RANKED STABLE MATCHING ON ANTIPODAL ORBITS

The residual sets are invariant under the antipodal involution  $X \mapsto -X$ . Since  $n > 0$ , no point in either set equals its negative; this is what it means here for the action of  $\{\pm 1\}$  to be free. Define the corresponding orbit sets

$$\bar{\mathcal{S}}_n = \mathcal{S}_n^\circ / \{\pm 1\}, \quad \bar{\mathcal{T}}_n = \mathcal{T}_n^\circ / \{\pm 1\}.$$

By (11), these sets have the same cardinality; write

$$|\bar{\mathcal{S}}_n| = |\bar{\mathcal{T}}_n| = m.$$

Each orbit is an antipodal pair  $[X] = \{X, -X\}$ . Choose the lexicographically smaller of  $X$  and  $-X$  as its canonical representative  $\text{pc}([X])$ . Let

$$G_n = K_{\bar{\mathcal{S}}_n, \bar{\mathcal{T}}_n}$$

be the complete bipartite graph on the two orbit sets.

Fix  $[E] \in \bar{\mathcal{S}}_n$  and  $[F] \in \bar{\mathcal{T}}_n$ , with canonical representatives  $E$  and  $F$ . Apply Lemma 2.4 to the two edges between chosen representatives  $E \leftrightarrow F$  and  $E \leftrightarrow -F$ . Let their direction keys be  $k_+(E, F)$  and  $k_-(E, F)$ .

**Lemma 4.1** (Edge ranking on antipodal orbits). *The two keys  $k_+(E, F)$  and  $k_-(E, F)$  are distinct. The quantity*

$$\kappa([E], [F]) = \min\{k_+(E, F), k_-(E, F)\}$$

*is independent of the chosen representatives. There is a unique sign*

$$\eta([E], [F]) \in \{\pm 1\}$$

*such that the representative-level edge*

$$E \longleftrightarrow \eta([E], [F])F$$

*has key  $\kappa([E], [F])$ .*

*Proof.* Write

$$\frac{E + F}{2} = qd, \quad \frac{E - F}{2} = ce$$

with primitive oriented directions  $d, e$ . Equality of the norms gives  $\langle d, e \rangle_Q = 0$ . The vectors are nonzero, so positive definiteness implies  $d \neq e$  and hence the keys are distinct. Replacing either representative by its negative interchanges the two signed midpoint lines, so their unordered pair of keys is unchanged. The smaller one determines the unique sign.  $\square$

*Example 4.2* (Choosing and lifting one signed orbit edge). Take the canonical representatives

$$E = (-2, -5, -2), \quad F = (-4, -3, 0)$$

of the orbits later denoted by  $s_1$  and  $t_1$ . There are two representative-level choices. For  $E \leftrightarrow F$ , the midpoint is

$$\frac{E + F}{2} = (-3, -4, -1) = -(3, 4, 1),$$

so its oriented primitive direction has key  $(36, 3, 4, 1)$ . For  $E \leftrightarrow -F$ , the midpoint is

$$\frac{E - F}{2} = (1, -1, -1),$$

whose key is  $(5, 1, -1, -1)$ . The second key is smaller. Therefore

$$\kappa([E], [F]) = (5, 1, -1, -1), \quad \eta([E], [F]) = -1,$$

and the canonical source representative lifts to

$$E \mapsto -F = (4, 3, 0).$$

For the other source point  $-E$ , multiplying both endpoints by  $-1$  gives the other lift  $-E \mapsto F$ . Thus the orbit edge encodes both point-level edges and their inverse signs.

If the actual source is  $\sigma E$ , where  $\sigma \in \{\pm 1\}$ , the lift of an orbit edge is

$$(12) \quad \sigma E \mapsto \sigma \eta([E], [F])F.$$

The same rule, read from the target, recovers the source.

**Proposition 4.3** (Equal-key classes are matchings). *For a fixed key  $k$ , the edges of  $G_n$  with key  $k$  form a matching: no two share a source or target orbit.*

*Proof.* At a fixed signed endpoint, a primitive direction determines at most one opposite signed endpoint by Lemma 2.4. It remains to check that the two signs of an endpoint do not create two orbits. If a fixed direction sends  $F$  to  $E = -F + 2qd$ , then at the endpoint  $-F$  the multiplier is  $-q$  and the partner is

$$-(-F) + 2(-q)d = -E.$$

Thus changing the sign of one endpoint changes the sign of the other and does not change its orbit. Hence a fixed key occurs at most once at either endpoint of  $G_n$ .  $\square$

*Example 4.4* (Two disjoint edges with the same key). The key

$$(5, 1, -1, -1)$$

occurs on the edge  $(s_1, t_1)$  computed in Example 4.2. For a second edge, take

$$E' = (-2, 5, -2) = \text{pc}(s_3), \quad F' = (0, -3, 4) = \text{pc}(t_4).$$

Their midpoint is  $(-1, 1, 1) = -(1, -1, -1)$ , so this edge has the same key. The sources  $s_1, s_3$  and targets  $t_1, t_4$  are distinct. Hence the two equal-key edges are disjoint, exactly as Proposition 4.3 requires.

Every vertex ranks its incident edges by increasing  $\kappa$ . Proposition 4.3 implies that equal-key edges are vertex-disjoint, so every vertex has a strict preference order. A pair  $(s, t) \notin M$  *blocks* a matching  $M$  if both  $s$  and  $t$  prefer each other to their partners in  $M$ , with being unmatched least preferred. A matching with no blocking pair is *stable*. Thus  $G_n$  is a stable-marriage instance with globally ranked pairs  $[4, 5, 1]$ : the two preference systems are induced by one global ranking of edges.

**Proposition 4.5** (Unique stable matching). *The graph  $G_n$  has a unique stable matching. It is obtained by processing the edge keys in increasing order and selecting every edge whose two endpoints are still unmatched. Since  $|\mathcal{S}_n| = |\mathcal{T}_n|$ , this stable matching is perfect.*

*Proof.* First run the stated greedy procedure. If an edge is not selected, then at least one of its endpoints was already matched when that edge was processed. The selected edge at that endpoint has smaller key; equality is impossible by Proposition 4.3. Thus an unselected edge cannot block, and the greedy matching is stable. It is also perfect: otherwise balance would leave an unmatched source and an unmatched target, but the edge joining them would have been selected when it was processed.

It remains to prove uniqueness. Consider the least key among the edges whose endpoints have not already been removed. Every edge of that key is forced in any stable matching: neither endpoint has a smaller edge in the remaining graph, so omission would make the two endpoints a blocking pair. Proposition 4.3 says that these edges have disjoint endpoints, and hence they can be selected and removed simultaneously. A removed endpoint prefers its selected edge to every edge processed later, and the restriction of a stable matching to the remaining vertices is stable. Induction therefore shows that every stable matching agrees with the greedy matching.  $\square$

*Example 4.6* (A two-by-two globally ranked instance). Before returning to the arithmetic graph, consider two sources  $s_1, s_2$  and two targets  $t_1, t_2$ . Give

the four edges the global ranks

|       | $t_1$ | $t_2$ |
|-------|-------|-------|
| $s_1$ | 1     | 3     |
| $s_2$ | 2     | 4     |

with smaller ranks preferred. The greedy rule first selects  $(s_1, t_1)$ . The edges of ranks 2 and 3 then meet an already matched vertex, so the next selected edge is  $(s_2, t_2)$ . Hence

$$M = \{(s_1, t_1), (s_2, t_2)\}.$$

The only other perfect matching is not stable, because  $(s_1, t_1)$  blocks it. This is the simplest instance of Proposition 4.5.

## 5. ASSEMBLY OF THE TWO-TO-ONE MAP

Let

$$\overline{M}_n : \overline{\mathcal{S}}_n \xrightarrow{\sim} \overline{\mathcal{T}}_n$$

be the unique stable matching of Proposition 4.5. Lift it to residual points using (12), and transfer back to the original coordinates. Denote the resulting residual bijection by

$$\Psi_n^\circ : \mathcal{B}_{n,\circ}^{\text{odd}} \xrightarrow{\sim} \mathcal{A}_{n,\circ}.$$

Define the odd-branch bijection

$$\Psi_n(X) = \begin{cases} \Lambda_n(X), & X \in D_n, \\ \Psi_n^\circ(X), & X \in \mathcal{B}_{n,\circ}^{\text{odd}}. \end{cases}$$

*Proof of Theorem 1.2.* The direct layer is a bijection  $D_n \rightarrow I_n$  by Theorem 3.2. The residual orbit matching is perfect by Proposition 4.5, and the sign lift pairs the two points in each source orbit with the two points in the matched target orbit. Hence  $\Psi_n$  is a bijection  $\mathcal{B}_n^{\text{odd}} \rightarrow \mathcal{A}_n$ .

Define

$$\Phi_n(x, y, z) = \begin{cases} (x, y, z/2), & z \text{ even}, \\ \Psi_n(x, y, z), & z \text{ odd}. \end{cases}$$

For  $(u, v, w) \in \mathcal{A}_n$ , the even branch contributes the preimage  $(u, v, 2w)$ , and the odd branch contributes the unique preimage  $\Psi_n^{-1}(u, v, w)$ . They are distinct because their third coordinates have opposite parity, and there are no further preimages on either branch. Thus every fiber has cardinality two.  $\square$

*Proof of Corollary 1.3.* For an odd squarefree congruent number  $n$ , Tunnell's theorem gives (3), so Theorem 1.2 applies.  $\square$

*Remark 5.1* (Role of cardinality). The formulas defining the direct layer and edge keys make sense without (3). The equality is used only to make the two residual orbit sets equally large; Proposition 4.5 then makes their stable matching perfect. The later generator and deferred-acceptance procedure compute this already defined matching.

*Example 5.2* (The direct  $n = 41$  fiber). Let  $Y = (2, -1, 1) \in \mathcal{A}_{41}$ . The even branch gives

$$(2, -1, 2) \mapsto (2, -1, 1),$$

while Example 3.3 gives the odd preimage

$$(-4, -1, 1) \mapsto (2, -1, 1).$$

Their last coordinates have opposite parity, so

$$\Phi_{41}^{-1}(2, -1, 1) = \{(2, -1, 2), (-4, -1, 1)\}.$$

This is the complete fiber in the direct lane announced in the introduction.

## 6. ARITHMETIC GENERATION OF PREFERENCE LISTS

The main theorem is now proved. The remaining sections make its residual matching computable without storing all  $m^2$  edges. We first bound the direction search and then generate one source's neighbors in preference order from a rank-two affine lattice.

**Lemma 6.1** (Bound for the preferred midpoint generator). *If  $d$  is the direction realizing the smaller signed orientation of a residual orbit edge, then*

$$Q(d) \leq \frac{n-1}{2}.$$

*Proof.* With notation as in the proof of Lemma 4.1, orthogonality and the two midpoint decompositions give

$$E = qd + ce, \quad F = qd - ce.$$

Therefore

$$n = q^2 Q(d) + c^2 Q(e).$$

The chosen orientation gives  $\text{key}(d) < \text{key}(e)$ , hence  $Q(d) \leq Q(e)$ . Since  $q$  and  $c$  are nonzero,

$$n \geq Q(d) + Q(e) \geq 2Q(d).$$

The claim follows because  $n$  is odd. □

*Example 6.2* (The midpoint bound is attained). In Example 2.5, the preferred direction satisfies

$$Q(d) = 20 = \frac{41-1}{2}.$$

Thus the bound in Lemma 6.1 can be an equality.

**6.1. An orthogonal-complement lattice.** Define the following isometric embedding of lattices:

$$\tau(X_1, X_2, X_3) = (X_1 + X_3, X_2, X_1 - X_3).$$

Then

$$\|\tau(X)\|^2 = Q(X), \quad \tau(X) \cdot \tau(Y) = \langle X, Y \rangle_Q.$$

The image of  $\tau$  is the index-two lattice

(13)

$$\mathcal{L}_\tau = \{r \in \mathbb{Z}^3 : r_1 \equiv r_3 \pmod{2}\}, \quad \tau^{-1}(r) = \left(\frac{r_1 + r_3}{2}, r_2, \frac{r_1 - r_3}{2}\right).$$

Fix a residual source orbit, let  $E$  be its canonical lifted representative, and put  $p = \tau(E)$ . Since  $\|p\|^2 = n$  and  $n$  is squarefree,  $p$  is primitive: a common divisor of its coordinates would have its square dividing  $n$ .

The geometric search is simple to state before introducing coordinates. If  $h = Q(d)$  and  $t = qh$ , then  $r = \tau(d)$  must be an integer point in

$$(14) \quad \underbrace{\{r : p \cdot r = t\}}_{\text{an affine plane}} \cap \underbrace{\{r : \|r\|^2 = h\}}_{\text{a sphere}} \cap \mathcal{L}_\tau.$$

The integer points parallel to the plane form the rank-two lattice  $p^\perp \cap \mathbb{Z}^3$ . The next lemma chooses coordinates on that lattice; the following subsection turns the plane-sphere intersection into a quadratic equation in two integers.

**Lemma 6.3** (Oriented kernel basis and Bézout vector). *There are deterministically computable vectors  $e_1, e_2, z_0 \in \mathbb{Z}^3$  such that*

$$e_1 \times e_2 = p, \quad p \cdot z_0 = 1.$$

*The pair  $e_1, e_2$  is an oriented basis of  $p^\perp \cap \mathbb{Z}^3$  and may be Lagrange–Gauss reduced while preserving  $e_1 \times e_2 = p$ .*

*Proof.* Primitivity of  $p$  gives a Bézout vector  $z_0$  with  $p \cdot z_0 = 1$ . A Smith-normal-form computation for the row vector  $p$  gives an integral basis  $e_1, e_2$  of its kernel. Every  $x \in \mathbb{Z}^3$  then has the unique decomposition

$$x = (x - (p \cdot x)z_0) + (p \cdot x)z_0.$$

Thus  $e_1, e_2, z_0$  form a basis of  $\mathbb{Z}^3$ . Since  $e_1 \times e_2$  is an integral multiple of the primitive normal vector  $p$ , its scalar product with  $z_0$  shows that this multiple is  $\pm 1$ . Reversing the kernel basis if necessary gives  $e_1 \times e_2 = p$ . The usual integral Lagrange–Gauss steps

$$(e_1, e_2) \mapsto (e_1, e_2 - me_1), \quad (e_1, e_2) \mapsto (e_2, -e_1)$$

have determinant 1, preserve the cross product, and terminate with  $|2e_1 \cdot e_2| \leq \|e_1\|^2 \leq \|e_2\|^2$ .  $\square$

For completeness, Smith normal form here means applying the Euclidean algorithm through integer column operations until the primitive row  $p$  is changed to  $(1, 0, 0)$ ; transforming the last two standard basis vectors back gives a basis of  $p^\perp \cap \mathbb{Z}^3$ . We use nonnegative Euclidean remainders and

lexicographic tie breaking, choose the corresponding Bézout vector, and use the nearest-integer Lagrange–Gauss step with ties rounded toward zero; if the two reduced vectors have equal norm, the lexicographically smaller oriented basis is chosen. These conventions make the auxiliary frame deterministic. The final ordered list is intrinsic, since its records are sorted by the direction key and do not depend on the chosen frame. These standard lattice algorithms are discussed, for example, in [3, Chap. 2].

Put

$$\begin{aligned}\alpha &= \|e_1\|^2, & \beta &= e_1 \cdot e_2, & \gamma &= \|e_2\|^2, \\ \delta &= z_0 \cdot e_1, & \varepsilon &= z_0 \cdot e_2, & \zeta &= \|z_0\|^2.\end{aligned}$$

Then

$$(15) \quad \alpha\gamma - \beta^2 = n.$$

Indeed,

$$\alpha\gamma - \beta^2 = \|e_1 \times e_2\|^2 = \|p\|^2 = n.$$

**6.2. Affine norm equations.** Suppose an incident representative-level edge has midpoint data

$$\frac{E+F}{2} = qd, \quad h = Q(d), \quad r = \tau(d).$$

Then

$$(16) \quad \|r\|^2 = h, \quad p \cdot r = qh.$$

Set  $t = qh$ .

**Lemma 6.4** (Parametrization of a lattice coset). *Every integral solution of  $p \cdot r = t$  is uniquely of the form*

$$r = tz_0 + \mu e_1 + \nu e_2, \quad \mu, \nu \in \mathbb{Z}.$$

For fixed  $h, q, \nu$ , the condition  $\|r\|^2 = h$  is equivalent to

$$(17) \quad \alpha\mu^2 + 2(\beta\nu + t\delta)\mu + \gamma\nu^2 + 2t\varepsilon\nu + t^2\zeta - h = 0.$$

*Proof.* The difference  $r - tz_0$  is orthogonal to  $p$ , and  $e_1, e_2$  form an integral basis of the orthogonal-complement lattice. Expanding the squared norm gives (17).  $\square$

Let

$$K = \beta\delta - \alpha\varepsilon, \quad M = h(n - q^2h).$$

The discriminant of (17), divided by 4, is

$$\Delta_\nu = (\beta\nu + t\delta)^2 - \alpha(\gamma\nu^2 + 2t\varepsilon\nu + t^2\zeta - h).$$

**Lemma 6.5** (Short interval identity). *One has*

$$(18) \quad n\Delta_\nu = \alpha M - (n\nu - tK)^2.$$

Put  $R = \lfloor \sqrt{\alpha M} \rfloor$ . By (18),  $\Delta_\nu \geq 0$  exactly when  $\nu$  lies in the interval

$$(19) \quad \left\lceil \frac{tK - R}{n} \right\rceil \leq \nu \leq \left\lfloor \frac{tK + R}{n} \right\rfloor.$$

For each such  $\nu$ , let  $s = \lfloor \sqrt{\Delta_\nu} \rfloor$ . If  $s^2 = \Delta_\nu$ , the possible values of  $\mu$  form the set

$$(20) \quad \left\{ \frac{-(\beta\nu + t\delta) + u}{\alpha} : u \in \{s, -s\}, \quad \alpha \mid -(\beta\nu + t\delta) + u \right\}.$$

If  $s^2 \neq \Delta_\nu$ , there are no roots. In particular, when  $\Delta_\nu = 0$ , the set (20) contains at most one integer rather than two copies of the same root.

*Proof.* The Gram matrix of the basis  $e_1, e_2, z_0$  is

$$G = \begin{pmatrix} \alpha & \beta & \delta \\ \beta & \gamma & \varepsilon \\ \delta & \varepsilon & \zeta \end{pmatrix}.$$

Since  $\det[e_1 \ e_2 \ z_0] = 1$ , one has  $\det G = 1$ . Expanding gives

$$n\zeta - (\alpha\varepsilon^2 - 2\beta\delta\varepsilon + \gamma\delta^2) = 1,$$

and a direct rearrangement using  $n = \alpha\gamma - \beta^2$  gives

$$K^2 + n(\delta^2 - \alpha\zeta) = -\alpha.$$

Now expand the discriminant as

$$\Delta_\nu = -n\nu^2 + 2tK\nu + t^2(\delta^2 - \alpha\zeta) + \alpha h.$$

Multiplying by  $n$ , applying the preceding identity, and using  $t = qh$  gives

$$n\Delta_\nu = \alpha(nh - t^2) - (n\nu - tK)^2 = \alpha h(n - q^2h) - (n\nu - tK)^2,$$

which is (18). Since  $n\nu - tK$  is integral, its square is at most  $\alpha M$  precisely when  $|n\nu - tK| \leq \lfloor \sqrt{\alpha M} \rfloor$ ; this is equivalent to (19). The quadratic formula gives (20), interpreted as a set.  $\square$

*Example 6.6* (Recovering one  $n = 41$  edge arithmetically). Return to  $E = (-2, -5, 2)$  and  $F = (4, -3, 0)$  from Example 2.5. The isometry gives

$$p = \tau(E) = (0, -5, -4).$$

One convenient oriented kernel basis and Bézout vector are

$$e_1 = (1, 0, 0), \quad e_2 = (0, -4, 5), \quad z_0 = (0, -1, 1).$$

These vectors can be found without treating them as a guess. The kernel equation  $p \cdot r = 0$  is  $-5r_2 - 4r_3 = 0$ , so  $r_2 = -4k$ ,  $r_3 = 5k$ , while  $r_1$  is free; this gives  $e_1, e_2$ . The Bézout identity  $5 - 4 = 1$  gives  $z_0$ . Thus  $e_1 \times e_2 = p$  and  $p \cdot z_0 = 1$ . For this frame,

$$\begin{aligned} \alpha &= 1, & \beta &= 0, & \gamma &= 41, \\ \delta &= 0, & \varepsilon &= 9, & \zeta &= 2, & K &= -9. \end{aligned}$$



The midpoint direction is  $d = (1, -4, 1)$ , so

$$h = Q(d) = 20, \quad q = 1, \quad t = 20, \quad M = 20(41 - 20) = 420.$$

Since  $R = \lfloor \sqrt{420} \rfloor = 20$  and  $tK = -180$ , the interval (19) becomes

$$\left\lceil \frac{-200}{41} \right\rceil \leq \nu \leq \left\lfloor \frac{-160}{41} \right\rfloor,$$

which leaves only  $\nu = -4$ . The short interval identity then gives

$$41\Delta_{-4} = 420 - (41(-4) - 20(-9))^2 = 164, \quad \Delta_{-4} = 4.$$

Thus (20) gives  $\mu = \pm 2$ . The choice  $\mu = 2$  produces

$$r = 20z_0 + 2e_1 - 4e_2 = (2, -4, 0), \quad \tau^{-1}(r) = (1, -4, 1) = d.$$

The other root gives a direction with negative first coordinate and is removed by the orientation convention. It is not lost: the outer loop also uses negative  $q$ , and the canonically oriented opposite direction is recovered in that loop when it yields a preferred signed edge. Finally,

$$-E + 2d = (4, -3, 0) = F.$$

Since the canonical representative of  $[F]$  is  $F_0 = (-4, 3, 0)$ , the generator records

$$((20, 1, -4, 1), [F_0], -1).$$

This is one complete pass through the arithmetic used to produce an entry of a preference list.

**6.3. The ordered-adjacency generator.** We now package the preceding calculation as a finite generator.

**Algorithm 6.7** (Ordered-adjacency generator). For a fixed residual source orbit  $[E]$ , with  $E = \text{pc}([E])$ , perform the following finite procedure:

- (1) let  $h = 1, 2, \dots, (n-1)/2$ ;
- (2) let  $q$  range over the nonzero odd integers satisfying  $q^2h < n$ , and put  $t = qh$  and  $M = h(n - q^2h)$ ;
- (3) let  $\nu$  range over the interval (19);
- (4) compute  $\Delta_\nu$  and the set of integral roots (20); for each root form  $r = tz_0 + \mu e_1 + \nu e_2$ ;
- (5) retain  $r$  only if  $r_1 \equiv r_3 \pmod{2}$  and  $d = \tau^{-1}(r)$  is primitive, canonically oriented, and has odd third coordinate;
- (6) form  $F = -E + 2qd$  and retain it only if  $F \in \mathcal{T}_n^\circ$ ; write  $F = \eta F_0$ , where  $F_0 = \text{pc}([F])$  and  $\eta \in \{\pm 1\}$ ;
- (7) compare  $d$  with the primitive oriented direction of  $(E - F)/2$ . Retain the record

$$(\text{key}(d), [F_0], \eta)$$

precisely when  $d$  has the smaller key.

For each  $h$ , insert the surviving records into a finite set and sort the set by  $(d_1, d_2, d_3)$ . Since the outer loop has increasing  $h$ , the resulting list is the strict preference list of  $[E]$ . All square roots and divisibility tests in the procedure are integer operations; in particular, the two signs in the quadratic formula are treated as a set. Viewed as a stateful generator, the procedure stores one sorted  $h$ -block at a time. After returning its records, it discards the block and advances to the next nonempty block.

**Theorem 6.8** (Correctness of the ordered-adjacency generator). *For every residual source orbit  $[E]$ , Algorithm 6.7 terminates and outputs every incident edge exactly once, in strictly increasing order of its key.*

*Proof.* Every retained record satisfies  $Q(d) = h$  and  $\langle E, d \rangle_Q = qh$ . Therefore

$$Q(-E + 2qd) = Q(E) - 4q \langle E, d \rangle_Q + 4q^2 Q(d) = n.$$

The condition  $F \in \mathcal{T}_n^\circ$  supplies the required congruence and exclusion conditions. Thus the record gives a valid edge between the chosen representatives. The final comparison keeps exactly the preferred one of the two signs of its target orbit, and  $\eta$  is exactly the sign required in (12).

Conversely, take an incident edge and choose its preferred signed lift from  $E$ . Lemma 6.1 gives the stated range for  $h = Q(d)$ . Its vectors  $p$  and  $r = \tau(d)$  satisfy (16). By Lemma 6.4,  $r$  has unique integral coordinates  $\mu, \nu$ , and Lemma 6.5 shows that the procedure recovers them. The  $\tau$ -lattice, primitivity, orientation, parity, and residual tests all hold for this lift, and the final comparison retains it.

Finally,  $h$  and  $d$  are intrinsic to a retained edge, while  $q = \langle E, d \rangle_Q / Q(d)$  is then forced. The affine coordinates  $\mu, \nu$  are unique, and (20) is a set even when  $\Delta_\nu = 0$ . Hence no record is emitted twice.  $\square$

*Example 6.9* (A complete generated preference row at  $n = 41$ ). For  $E = (-2, -5, 2)$ , direct enumeration of the residual target set gives four target orbits. Write their canonical representatives as

$$\begin{aligned} F_1 &= (-4, -3, 0), & F_2 &= (0, -3, 4), \\ F_3 &= (-4, 3, 0), & F_4 &= (0, -3, -4). \end{aligned}$$

Running Algorithm 6.7 gives the following surviving signed midpoint for each orbit. The last column records the  $Q$ -value of the discarded midpoint direction obtained from the opposite sign.

| orbit   | $\eta$ | $(E + \eta F_i)/2$ | $\kappa([E], [F_i])$ | other $Q$ |
|---------|--------|--------------------|----------------------|-----------|
| $[F_1]$ | -1     | $(1, -1, 1)$       | $(5, 1, -1, 1)$      | 36        |
| $[F_2]$ | -1     | $(-1, -1, -1)$     | $(5, 1, 1, 1)$       | 36        |
| $[F_3]$ | -1     | $(1, -4, 1)$       | $(20, 1, -4, 1)$     | 21        |
| $[F_4]$ | 1      | $(-1, -4, -1)$     | $(20, 1, 4, 1)$      | 21        |

The corresponding affine data  $(h, q, \mu, \nu)$ , in the same row order, are

$$(5, 1, 2, -1), \quad (5, -1, 2, 1), \quad (20, 1, 2, -4), \quad (20, -1, 2, 4).$$

For example, the third row is the full calculation in Example 6.6; its opposite sign gives the direction  $(3, 1, -1)$  of  $Q$ -value 21. In the second and fourth rows the displayed midpoint is the negative of its canonically oriented primitive direction, which is why the key has positive first coordinate. Sorting the surviving records gives the entire strict preference row

$$[F_1], [F_2], [F_3], [F_4]$$

with keys in the order displayed. Thus the example shows one complete list, including the four opposite-sign candidates rejected by the final comparison, rather than only one successful record.

*Remark 6.10* (Reduced-basis search-interval bound). For a Lagrange–Gauss reduced basis,

$$|2\beta| \leq \alpha \leq \gamma.$$

Equation (15) then gives  $\alpha \leq 2\sqrt{n/3}$ . Thus the interval (19) is shorter than the corresponding unrestricted scan of the orthogonal-complement lattice. This observation is not used in the correctness or matching-phase proofs.

## 7. DEFERRED ACCEPTANCE WITH GENERATED PREFERENCES

Proposition 4.5 describes the matching by a conceptual global greedy procedure. Sorting all edge keys at once, however, would defeat the purpose of generating preference lists one at a time. We instead use the source-proposing Gale–Shapley algorithm [4]. Here an iterator is a stateful generator for a fixed preference list; laziness affects only when an entry is computed, not the underlying preference order.

Enumerate the finite residual orbit sets  $\overline{\mathcal{S}}_n$  and  $\overline{\mathcal{T}}_n$ , and attach to each source  $s \in \overline{\mathcal{S}}_n$  the ordered generator of Theorem 6.8. A target compares two proposals by their global edge rank.

**Algorithm 7.1** (Source-proposing deferred acceptance with generated preferences). Initially every source is free and every target tentatively holds no proposal. Place the sources in a first-in, first-out queue in lexicographic order of their canonical representatives. Repeatedly remove a free source  $s$  from the front of the queue and request the next record  $(k, t, \eta)$  from its ordered generator.

- (1) If  $t$  tentatively holds no proposal, it tentatively holds  $(s, k, \eta)$ .
- (2) If  $t$  tentatively holds  $(s', k', \eta')$  and  $k < k'$ , then  $t$  replaces that record by  $(s, k, \eta)$  and  $s'$  is appended to the queue.
- (3) Otherwise  $t$  rejects  $s$ , which is appended to the queue.

The algorithm ends when the queue is empty.

*Example 7.2* (Deferred acceptance on two sources). Run Algorithm 7.1 on the ranks in Example 4.6, with initial queue  $(s_1, s_2)$ . The proposals are

$$s_1 \longrightarrow t_1, \quad s_2 \longrightarrow t_1 \text{ (rejected)}, \quad s_2 \longrightarrow t_2.$$

The first proposal is held by  $t_1$ . The second has rank 2, so  $t_1$  keeps its rank-1 proposal from  $s_1$  and rejects  $s_2$ . The queued source then proposes to  $t_2$ . The queue is empty and the output is  $\{(s_1, t_1), (s_2, t_2)\}$ , the unique stable matching found by the greedy argument.

**Theorem 7.3** (Correctness of deferred acceptance). *Algorithm 7.1 terminates and returns the unique perfect stable matching of Proposition 4.5. The stored sign  $\eta$  lifts every selected edge to a representative-level source-target pair. The inverse at a target is recovered by rerunning the same deterministic global computation; no proposal transcript is required.*

*Proof.* A source proposes in strict preference order and never proposes to the same target twice. We first check that a free source never exhausts its generator. If a free source  $s$  had proposed to all  $m$  targets, then every target would hold a proposal: once a target receives its first proposal, it remains occupied thereafter. The held proposals would come from  $m$  distinct sources, since a source can be held by at most one target and proposes again only after becoming free. None could come from the free source  $s$ , leaving only  $m - 1$  possible sources, a contradiction. Thus every request made by the algorithm has a next record. Since there are only  $m^2$  possible proposals, the algorithm terminates.

When the queue is empty, every source is held by a target. There are  $m$  sources and  $m$  targets, and no target holds two sources, so the resulting matching is perfect.

Suppose  $(s, t)$  blocked the final matching. Since  $s$  prefers  $t$  to its final partner,  $s$  proposed to  $t$  earlier. The target either rejected  $s$  immediately or later displaced it for a proposal of smaller key. A target's held key only decreases, so its final edge is smaller than  $(s, t)$ , a contradiction. The output is stable and hence equals the unique stable matching of Proposition 4.5.

For the inverse, let  $F_0$  be the canonical representative of a matched target orbit, and suppose its matched record is  $(E_0, F_0, \eta)$  with  $E_0$  canonical. The lifted matching sends  $E_0$  to  $\eta F_0$ . Hence an actual target  $\rho F_0$ , with  $\rho \in \{\pm 1\}$ , has preimage  $\rho \eta E_0$ . Lemma 4.1 makes  $\eta$  intrinsic to the two orbits, and rerunning the deterministic computation reconstructs the same matching and the same sign. This recovers the inverse at the given target without retaining the first run's proposal transcript.  $\square$

*Example 7.4* (Recovering a residual preimage). Take the actual lifted target  $F = (4, -3, 0)$ . Its canonical orbit representative is  $F_0 = (-4, 3, 0)$ , so  $F = \rho F_0$  with  $\rho = -1$ . Rerunning the deterministic matching returns the record from Example 6.9 with

$$E_0 = (-2, -5, 2), \quad \eta = -1.$$

The inverse sign formula gives

$$\rho \eta E_0 = (-1)(-1)E_0 = E_0.$$

Finally,  $\iota_B^{-1}(E_0) = (-2, -5, 1)$ . Thus the inverse at the original target is

$$(4, -3, 0) \mapsto (-2, -5, 1).$$

The example makes explicit that “recovering the inverse” means rerunning the global deterministic matching and then applying a local sign calculation; it is not a separate pointwise formula.

#### 8. THE TWO $n = 41$ EXAMPLE LANES

The direct lane was completed in Examples 3.3 and 5.2:

$$(-4, -1, 1) \mapsto (2, -1, 1) \quad \text{with even companion } (2, -1, 2).$$

We now complete the distinct residual lane

$$(-2, -5, 1) \mapsto (4, -3, 0) \quad \text{with even companion } (4, -3, 0).$$

The two odd-source mechanisms are alternatives on disjoint subsets; they are not successive steps in one fiber.

*Example 8.1* (The residual matching for  $n = 41$ ). The residual graph for  $n = 41$  has four source orbits  $s_1, \dots, s_4$  and four target orbits  $t_1, \dots, t_4$ . We index them by the following canonical lifted representatives:

$$\begin{aligned} \text{pc}(s_1) &= (-2, -5, -2), & \text{pc}(s_2) &= (-2, -5, 2), \\ \text{pc}(s_3) &= (-2, 5, -2), & \text{pc}(s_4) &= (-2, 5, 2), \\ \text{pc}(t_1) &= (-4, -3, 0), & \text{pc}(t_2) &= (-4, 3, 0), \\ \text{pc}(t_3) &= (0, -3, -4), & \text{pc}(t_4) &= (0, -3, 4). \end{aligned}$$

In the next display, each row is in preference order and the bracketed four-tuple is the corresponding edge key:

$$\begin{aligned} s_1 : & \quad t_1[5, 1, -1, -1], \quad t_3[5, 1, 1, -1], \quad t_2[20, 1, -4, -1], \quad t_4[20, 1, 4, -1], \\ s_2 : & \quad t_1[5, 1, -1, 1], \quad t_4[5, 1, 1, 1], \quad t_2[20, 1, -4, 1], \quad t_3[20, 1, 4, 1], \\ s_3 : & \quad t_4[5, 1, -1, -1], \quad t_2[5, 1, 1, -1], \quad t_3[20, 1, -4, -1], \quad t_1[20, 1, 4, -1], \\ s_4 : & \quad t_3[5, 1, -1, 1], \quad t_2[5, 1, 1, 1], \quad t_4[20, 1, -4, 1], \quad t_1[20, 1, 4, 1]. \end{aligned}$$

The complete generated row in Example 6.9 is exactly the  $s_2$  row here: its  $F_1, F_2, F_3, F_4$  are respectively  $t_1, t_4, t_2, t_3$ . Thus the abstract generator calculation and this graph table describe the same four edges. With the queue convention of Algorithm 7.1, the proposal sequence is

$$\begin{aligned} s_1 &\longrightarrow t_1, & s_2 &\longrightarrow t_1 \quad (\text{rejected}), & s_3 &\longrightarrow t_4, \\ s_4 &\longrightarrow t_3, & s_2 &\longrightarrow t_4 \quad (\text{rejected}), & s_2 &\longrightarrow t_2. \end{aligned}$$

Thus the algorithm makes six proposals. Its lifted residual pairs are

$$\begin{aligned} (-2, -5, -2) &\longleftrightarrow (4, 3, 0), \\ (-2, -5, 2) &\longleftrightarrow (4, -3, 0), \\ (-2, 5, -2) &\longleftrightarrow (0, -3, 4), \\ (-2, 5, 2) &\longleftrightarrow (0, -3, -4). \end{aligned}$$

For instance,

$$(-2, -5, 1) \in \mathcal{B}_{41}^{\text{odd}} \mapsto (4, -3, 0) \in \mathcal{A}_{41}.$$

Thus

$$\Phi_{41}^{-1}(4, -3, 0) = \{(-2, -5, 1), (4, -3, 0)\}.$$

## 9. PARKING FUNCTIONS AND MATCHING-PHASE BOUNDS

A sequence  $(a_1, \dots, a_m)$  of positive integers is a *parking function* if its nondecreasing rearrangement  $b_1 \leq \dots \leq b_m$  satisfies  $b_i \leq i$  for  $1 \leq i \leq m$ ; see [7, 8]. For  $s \in \overline{\mathcal{S}}_n$ , let  $p(s)$  be the number of proposals made by  $s$ . Equivalently,  $p(s)$  is the rank of its final target in its preference list. In the proof below,  $M = \overline{M}_n$  denotes the final stable matching.

**Theorem 9.1** (The proposal counts form a parking function). *Order the sources  $s_1, \dots, s_m$  so that the keys of their final matched edges are nondecreasing. Then*

$$p(s_i) \leq i \quad (1 \leq i \leq m).$$

*Let  $b_1 \leq \dots \leq b_m$  be the nondecreasing rearrangement of the proposal counts. Then  $b_i \leq i$ , so the proposal-count sequence is a parking function. Moreover,*

$$(21) \quad \sum_{s \in \overline{\mathcal{S}}_n} p(s) \leq \frac{m(m+1)}{2}.$$

*Proof.* Fix a source  $s$ , and let  $t$  be a target that  $s$  prefers to its final target. Stability forces

$$\kappa(M^{-1}(t), t) < \kappa(s, t) < \kappa(s, M(s));$$

otherwise  $(s, t)$  would block. Distinct preferred targets give distinct matched edges of key strictly smaller than the matched edge of  $s$ . Hence

$$p(s) - 1 \leq \#\{e \in M : \kappa(e) < \kappa(s, M(s))\}.$$

For  $s = s_i$ , the right-hand side is at most  $i - 1$ , proving  $p(s_i) \leq i$ . For each  $i$ , the first  $i$  sources in this ordering provide  $i$  proposal counts, all at most  $i$ . Hence the  $i$ th smallest proposal count is at most  $i$ , which is the parking-function condition. Summing the inequalities  $p(s_i) \leq i$  gives (21).  $\square$

*Example 9.2* (The parking function for  $n = 41$ ). The residual run makes the six proposals

$$s_1 \rightarrow t_1, \quad s_2 \rightarrow t_1, \quad s_3 \rightarrow t_4, \quad s_4 \rightarrow t_3, \quad s_2 \rightarrow t_4, \quad s_2 \rightarrow t_2.$$

Thus

$$p(s_1) = 1, \quad p(s_2) = 3, \quad p(s_3) = 1, \quad p(s_4) = 1.$$

Its nondecreasing rearrangement is  $(1, 1, 1, 3)$ , and

$$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 3 \leq 4.$$

Thus it is a parking function. The total number of proposals is 6, while Theorem 9.1 gives the bound  $4 \cdot 5/2 = 10$ . In the control layer this means six generator advances and six proposal-processing steps, of which exactly

two involve a key comparison against a held record, illustrating the count used in Theorem 9.5.

**Proposition 9.3** (Sharpness). *The bound (21) is sharp for the class of globally ranked-pairs instances, even when every key class consists of a single edge.*

*Proof.* Let the sides be  $\{s_1, \dots, s_m\}$  and  $\{t_1, \dots, t_m\}$ , and give  $(s_i, t_j)$  the distinct label  $(m+1)i + j$ . Every source orders the targets as  $t_1, t_2, \dots, t_m$ , and every target orders the sources as  $s_1, s_2, \dots, s_m$ . In increasing-label greedy order,  $(s_1, t_1)$  is selected first; after its endpoints are removed,  $(s_2, t_2)$  is selected, and induction gives the diagonal stable matching. The final target  $t_i$  has rank  $i$  in the list of  $s_i$ , so  $s_i$  makes exactly  $i$  proposals. The total is  $m(m+1)/2$ .  $\square$

*Example 9.4* (Sharpness with three sources). For  $m = 3$ , all sources rank the targets as  $t_1, t_2, t_3$ , and all targets rank the sources as  $s_1, s_2, s_3$ . Starting with the queue  $(s_1, s_2, s_3)$ , source  $s_1$  is held by  $t_1$ . Sources  $s_2$  and  $s_3$  are then rejected by  $t_1$ ; next  $s_2$  is held by  $t_2$ , and  $s_3$  is rejected by  $t_2$  before reaching  $t_3$ . The proposal counts are therefore

$$p(s_1) = 1, \quad p(s_2) = 2, \quad p(s_3) = 3,$$

whose total  $6 = 3 \cdot 4/2$  attains the bound.

**Theorem 9.5** (Matching-phase operation and auxiliary-storage bound). *Let*

$$N = \sum_{s \in \bar{\mathcal{S}}_n} p(s)$$

*be the number of proposals. Conditional on access to the ordered-adjacency generators of Theorem 6.8, the deferred-acceptance control layer makes  $N$  generator-advance calls, performs  $O(N)$  key comparisons and queue or assignment operations, and stores  $O(m)$  auxiliary records outside generator state. In particular,*

$$N \leq \frac{m(m+1)}{2}.$$

*The stated bound concerns the deferred-acceptance phase only. It does not include the arithmetic work or storage internal to the generators, the enumeration of the residual orbits, or integer bit complexity.*

|                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Counted by Theorem 9.5:</b> One generator advance for each proposal; key comparisons; queue and assignment updates; and source, target, and queue records outside the generators.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Not counted by Theorem 9.5:</b> Construction of the residual orbit sets; arithmetic used to produce each generator record; generator-internal storage; integer bit complexity; and the cost of rerunning the computation for an inverse query.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

*Proof.* Every proposal performs at most one key comparison—exactly one when the proposed target is already held, and none when it is free—and a constant number of queue and matching-array updates. Apart from the external generators, the retained state consists of the source status array, one held record per target, and the queue of free sources, all of size  $O(m)$ . The bound on  $N$  is Theorem 9.1.  $\square$

*Remark 9.6* (On possible subquadratic bounds). Proposition 9.3 shows that an  $O(m \log m)$  worst-case bound on the number of proposals cannot follow from the globally-ranked-pairs property alone. Such a bound for the Tunnell-form instances considered here would require additional arithmetic information about the distribution of their midpoint directions.

## 10. CONCLUDING REMARKS

The construction brings together a local arithmetic bijection and a global combinatorial completion. Quarter-turns account for congruence-defined subsets, while primitive midpoint directions rank the edges between the remaining orbits. This ranking forces both the unique stable matching and the parking-function bound. The examples for  $n = 41$  give one complete fiber of each odd-branch type: one using a quarter-turn and one using the residual stable matching. In both cases the universal even branch supplies the other preimage.

The ordered-adjacency generator avoids storing the complete  $m \times m$  edge table. It would be interesting to determine whether the arithmetic distribution of midpoint directions yields a better complexity bound for these Tunnell-form instances, or whether the residual partner admits a direct description from one representation.

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## DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the mathematical development and preparation of this work, the authors used OpenAI’s ChatGPT and Codex through an AI-assisted solver–referee pipeline for symbolic experimentation, literature searches, proof



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